Development of a Web Service for Creating and Managing an Interactive Digital Portfolio and Resume

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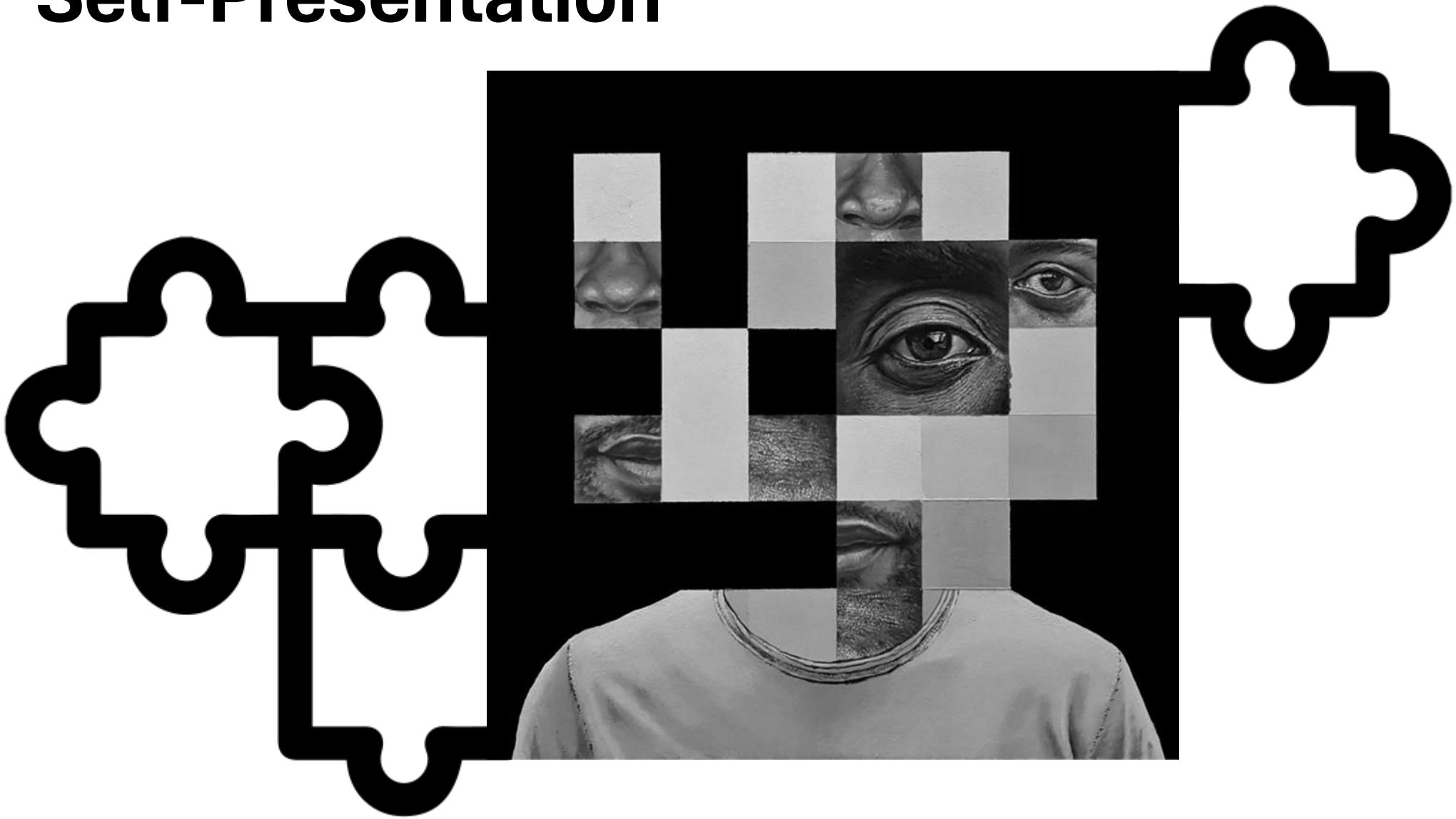
The Modern Challenge of Professional Self-Presentation



The Gap in Current Solutions

Category	Key Limitation
Professional Networks	No deep project visualization
Design Portfolios	Niche audience, no resume
Resume Builders	Static format, weak project display
Code Repositories	Code-focused, no visual portfolio or career analytics

LinkedIn GitHub Bēhance kickresume about.me

Comparison of Existing Solutions

Platform / Feature	Interactive Skill Visualization	Rich Project Showcase	Customizable Resume Templates	PDF Export	Public Profile Link	Data-Driven Skill Analysis
About.me	✕	✕	✕	✕	✓	✕
CakeResume / Kickresume	! (Basic)	! (Limited)	✓	✓	✓	✕
Behance / Dribbble	✕	✓ (For design)	✕	✕	✓	✕
LinkedIn	! (Simple list)	! (Text-based)	! (Limited)	! (Basic)	✓	! (Very basic)
GitHub Pages	✕	✓ (Code-focused)	✕	✕	✓	✕
My Project PortfolioPro	✓ (Charts, graphs)	✓ (Full media support)	✓ (Fully customizable)	✓ (Smart, styled)	✓ (With privacy control)	✓ (Core innovation)

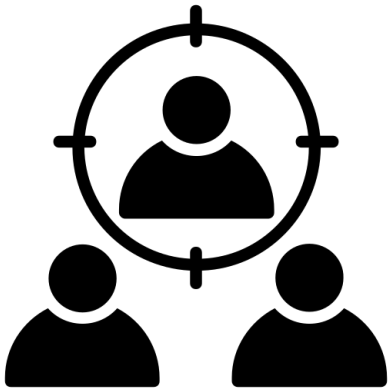
An All-in-One Interactive Portfolio

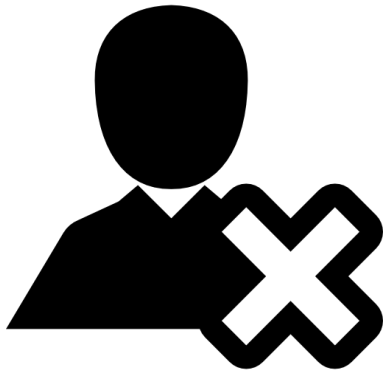
Key Features

- **Interactive Constructor** – drag & drop + visual skill charts
- **Smart PDF Export** – one-click polished resume
- **Public Profile** – shareable link, full design control
- **Analytics Module** – data-driven career insights (e.g., "Add React for +65% matches")

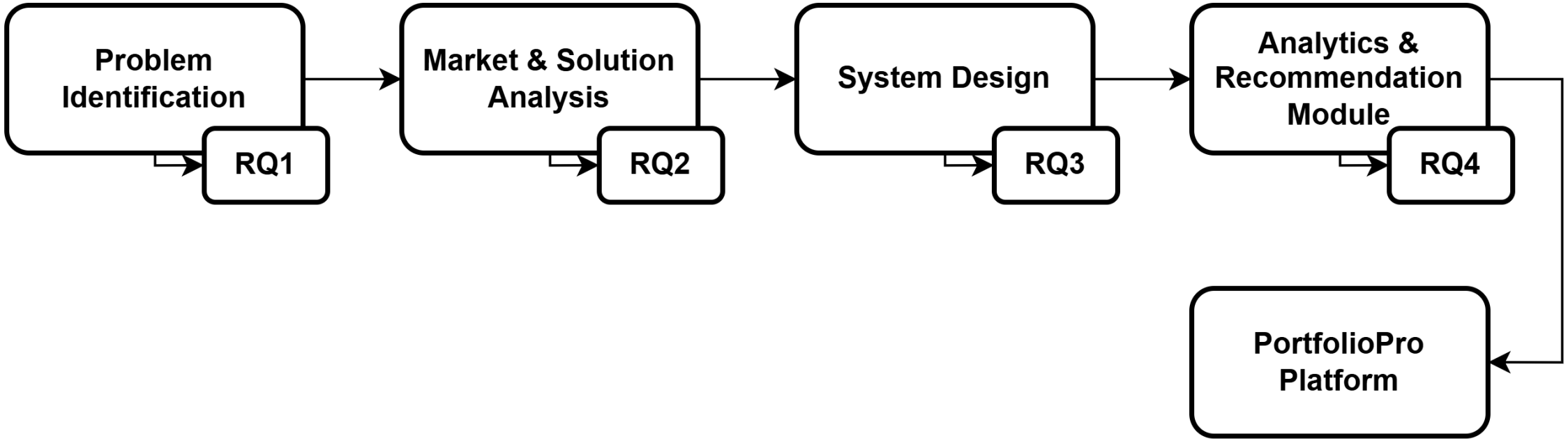
Limitations

- **Niche audience** – devs, designers
- **External API** – relies on HH.ru
- **Self-reported data** – not verified





Research Questions



Research Questions

RQ1: How does the design of skill visualization (charts, graphs) in a digital portfolio influence an employer's perception of a candidate?

RQ2: What design approaches in existing platforms successfully combine user-friendly creation tools with employer-oriented utility?

RQ3: What methods and approaches are most suitable for developing an all-in-one portfolio tool that combines key features from existing solutions?

RQ4: What methods and data allow for analysis of a user's skills and provide personalized recommendations for career growth?

RQ1: Visualization & Perception

Memorability

Visual embellishments increase memorability of charts. (Bateman et al., 2010)

Structured self-presentation boosts perceived credibility. (Toma & Hancock, 2011)

Accuracy

High memorability does not guarantee correct interpretation; embellishments may distort meaning (Borkin et al., 2013)

Risk: Over-stylized visuals can prioritize aesthetics over accurate communication.

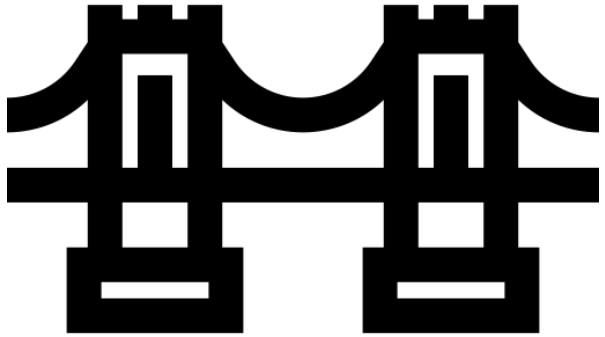
Key Insight :
Skill visualizations must balance engaging design with clear and interpretable information.

RQ2: Dual-Audience Design

Creator Perspective (Usability)

Usability guidelines show that simple forms reduce errors and increase efficiency. (Bargas-Avila et al., 2011)

Trust in the platform (transparency and reliability) encourages adoption by both users and employer (Gefen et al., 2008)



Employer Perspective (Evaluative Depth)

The structure of information affects assessment quality; oversimplification may reduce evaluation depth. (Langer et al., 2018)

Key Insight :
The system must be intuitive for creators while providing structured, trustworthy signals for employers.

RQ3: System Architecture

Agile Development

Agile = flexibility, iterative delivery, adaptability to change.
(Petersen & Wohlin, 2010)

Data-Intensive Architecture

Modular, scalable architecture needed for integrating user data + external APIs.
(Chen et al., 2012)

Key Insight :
Combine agile process with solid, modular backend to ensure both flexibility and robustness.

RQ4: Predictive Analytics

Descriptive (Static Analysis)

Big data from job vacancies reveals patterns in labor market demand.
(Frank et al., 2019)

Skills function as economic signals valued by the labor market.
(Deming, 2017)

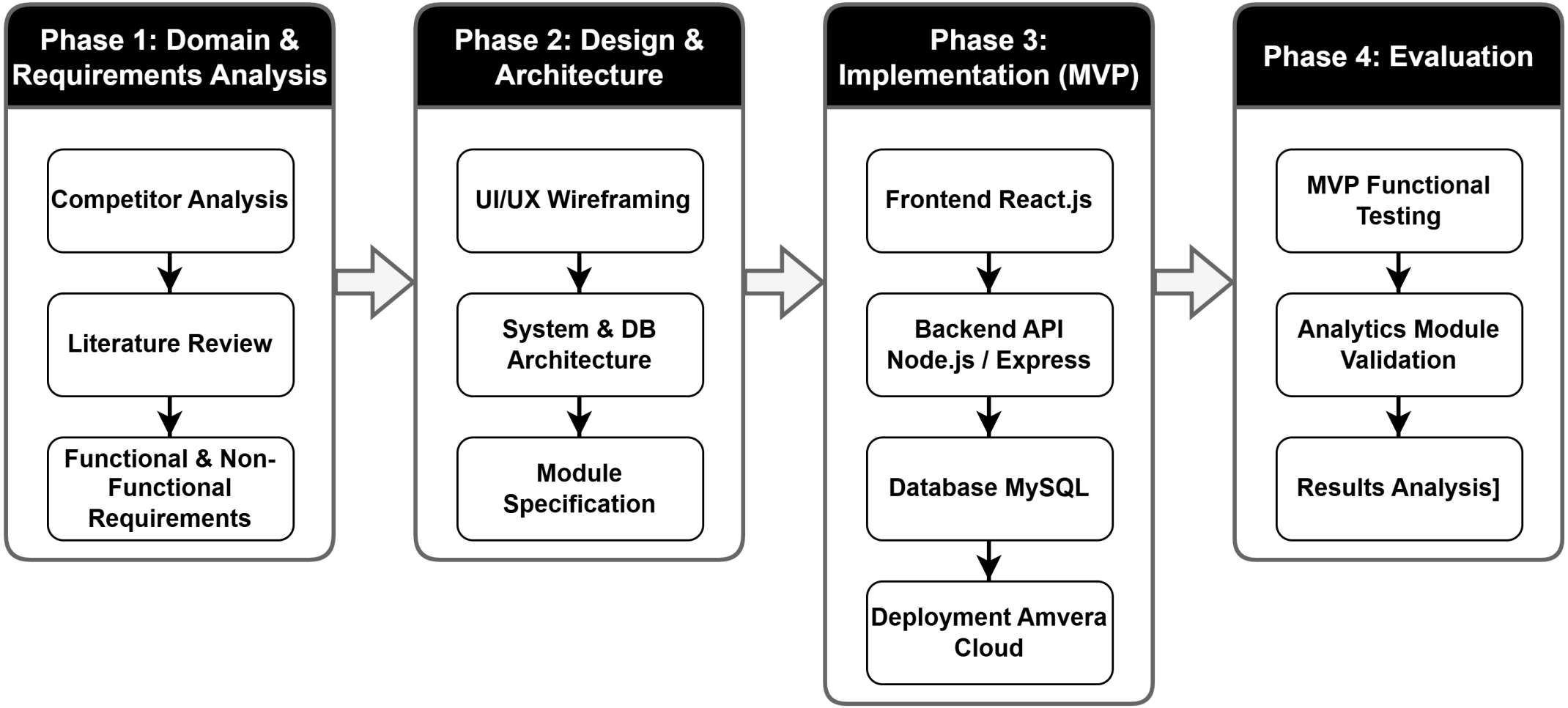
Predictive (Actionable Insights)

Predictive analytics estimates future trends and skill gaps.
(Shmueli & Koppius, 2011)

Goal: Convert labor-market data into personalized skill recommendations.

Key Insight :
Analyze vacancy data (HH.ru) to identify high-demand skills and compare them with user profiles to generate targeted recommendations.

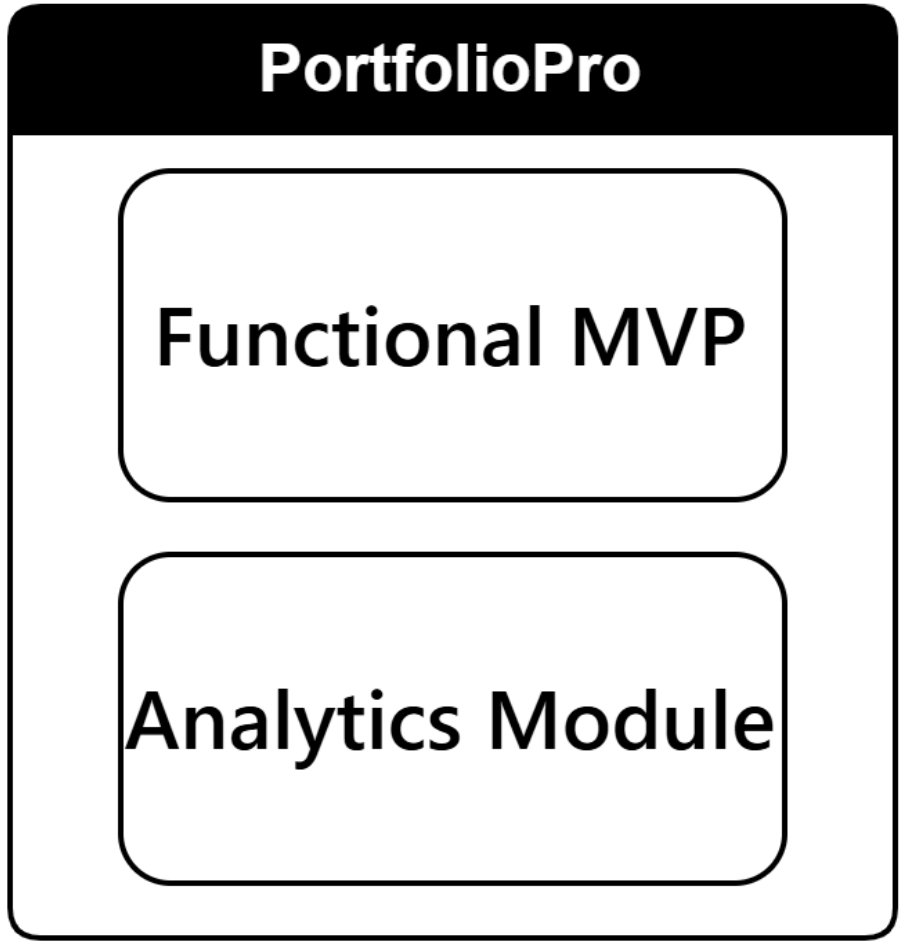
Methods


Phase 1: Domain & Requirements Analysis
Competitor Analysis
Literature Review
Functional & Non-Functional Requirements
Phase 2: Design & Architecture
UI/UX Wireframing
System & DB Architecture
Module Specification
Phase 3: Implementation (MVP)
Frontend React.js
Backend API Node.js / Express
Database MySQL
Deployment Amvera Cloud
Phase 4: Evaluation
MVP Functional Testing
Analytics Module Validation
Results Analysis]

Anticipated Results

A working platform with user authentication, interactive portfolio constructor, and one-click PDF resume generation.

Integration with HH.ru API for real-time skill-gap analysis and personalized career recommendations.



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